Department of Physics
Faculty of Science
University of Douala, Cameroon
Email: steve.teguia@facsciences-uy1.cm

June 20, 2026

Prof. Gregory Scholes
Editor-in-Chief
The Journal of Physical Chemistry Letters
American Chemical Society

Dear Prof. Scholes,

We submit the second major revision of our manuscript "Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems" (Manuscript ID: jz-2026-00994t.R2) for consideration as a Letter in *The Journal of Physical Chemistry Letters.*

We thank both reviewers for their continued, careful evaluation of our work. The revisions address all points raised in the second-round reports:

1. **Transfer yield redefined (Reviewer 3).** Φ_{FT} is now the long-time equilibrium population of BChl 3 (Eq. 7). The enhancement $\eta = (\Phi_{\mathrm{FT}}^{\mathrm{filt}} - \Phi_{\mathrm{FT}}^{\mathrm{broad}})/\Phi_{\mathrm{FT}}^{\mathrm{broad}}$ is physically transparent and directly reflects directed energy flow. Production value: $\eta = 0.39(4)$ (vibronic bath, $N = 200$).
2. **Vibronic-basis terminology clarified (Reviewer 2).** "Polaron-dressed states" replaced by "vibronically dressed states" throughout. The displaced-oscillator derivation is confined to the SI.
3. **Early vibronic-resonance literature cited (Reviewer 2).** The Introduction now discusses the 2015 and 2022 JPC Letters references.
4. **Spectral density corrected (Reviewer 3).** Fig. 3e displays $J(\omega)$ over 0–2000 $\mathrm{cm}^{-1} withtheDrude- -Lorentzbackgroundand12discretevibronicpeaks. \lambda_{\mathrm{total}} = 128\,\mathrm{cm}^{-1}$ is broken down in SI Table S1.
5. **Dissipation confirmed (Reviewer 3).** The production ensemble ($N = 200$, $L = 8$, $K = 2$, 12 vibronic modes) shows clear l_1-norm decay and population equilibration. $\eta = 0.39(4)$, a 2.2-fold increase over the previous Drude-only value.
6. **Computational jargon removed (Reviewer 3).** All hardware-specific language removed from the main text. Fig. 2 (resource flowchart) moved to SI Section S5.
7. **PT-HOPS scaling clarified (Reviewer 3).** The $\mathcal{O}(1)$ claim corrected; $K = 2$ convergence justified via γ_D vs. first Matsubara frequency (MAE $= 3.32 \times 10^{-5}$).
8. **Robustness analysis.** Comprehensive sweeps demonstrate robustness:
   - Temperature: η decreases from 0.543 (285 K) to 0.386 (295 K) — coherent mechanism confirmed.
   - Bath: $\eta \in [0.28, 0.62]$ for $\lambda, \gamma \pm 20\,\%$.
   - Filter: η ranges from -0.96 (off-resonant) to 0.65 (broadband).
9. **Proofreading completed.** The manuscript has been independently reviewed by two co-authors and an automated spell-checker.

These updates reinforce our finding that selective vibronic excitation enhances directed transfer by $\eta = 0.39$ at room temperature. We believe the revisions fully address all reviewer concerns.

Response to manuscript formatting request (28-Apr-2026).

1. **Section headings removed.** Major divisions are now indicated by bold run-in paragraph leads; subsections use unnumbered `\subsection*{}`.
2. **TOC Graphic included** via the `tocentry` environment.
3. **Supporting Information** contains no highlights, tracked changes, or colored text.

Cover art. We submit the TOC Graphic (`TOC_Graphic.png`) for supplementary cover consideration. The image depicts the dual-band filtering scheme applied to the FMO complex.

All authors have approved this revised manuscript. We declare no conflicts of interest. This work received no external funding.

Thank you for your continued consideration.

Sincerely,

Steve Cabrel Teguia Kouam

Steve Cabrel Teguia Kouam Steve Cabrel Teguia Kouam
Department of Physics
University of Douala, Cameroon
Email: steve.teguia@facsciences-uy1.cm